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Lingua inglese

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SMART CITY INFRASTRUCTURE

Tangible infrastructure (physical aspects): buildings, vehicles and layout that allow superior function

- it can optimise telecommunication, modern healthcare facilities, transportation networks and urban planning

Non-tangible infrastructure: digital networks and data

- this technological integration boosts sustainability and high quality of life

SMART CITY INFRASTRUCTURE

Core infrastructural elements:

- Internet of Things (IoT)
- Information & Communications Technology
- Automation
- Artificial Intelligence (AI)

SMART CITY INFRASTRUCTURE

Internet of Things (IoT)

- the network of physical devices, vehicles and appliances that carry smart technologies
- by interconnecting data from their sensors and software, “smart objects” help decision-makers optimize the overall functionality of a city
- the interconnectedness of smart objects ranges from home appliances to transportation systems

SMART CITY INFRASTRUCTURE

Information & Communications Technology (ICT)

- while IoT is the network of physical objects, ICT is the technological network that keeps smart cities functioning
- this owes to its processing and exchange of data, including all the other functionalities in between: storage, representation, security, management and more

SMART CITY INFRASTRUCTURE

Automation

- it mitigates the need for human activity on daily tasks
- smart cities use automation to respond quickly to real-time data, with uses like motion-activated streetlights
- automation leads to energy efficiency while also improving the quality of life for inhabitants

SMART CITY INFRASTRUCTURE

Artificial Intelligence (AI)

- Alongside automation, AI enhances smart cities by reducing dependence on moments of menial human decision-making
- AI algorithms can be used to optimize waste collection routes, or help law enforcement by analysing data from security cameras to detect crime

SMART CITY INFRASTRUCTURE

Technological applications that already exist in smart city infrastructure:

- **Smart grids:** optimize energy consumption and distribution, thus reducing waste and promoting renewable energy sources

- **Environmental monitoring systems:** use sensors and data analytics to track air and water quality, noise levels, and other environmental factors

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- **Smart transportation systems:** include automated vehicles, intelligent traffic management systems and electric vehicle charging infrastructure to reduce congestion and emissions

- **Blockchain technology**: boosts transparency in waste management, supply chains and energy grids

- **GIS (Geographic Information System)**: helps urban planners with integration of green spaces and alleviating urban heat island effects

(Source: https://www.ie.edu/uncover-ie/smart-city-infrastructure-ie-architecture-and-design/)

SMART CITIES

Smart city examples:

Singapore:

The global benchmark for data-driven urban planning. Singapore uses a network of over 50,000 sensors to monitor traffic flow, air quality, and crowd levels. It features autonomous shuttles, cashless digital IDs, and a "Digital Twin" of the city that allows urban planners to simulate and test infrastructure changes before building them.

SMART CITIES

Zurich, Switzerland:

Frequently ranked at the top of the IMD Smart City Index, Zurich places its citizens at the centre of its strategy. The city utilizes a heavily digitized, smart energy grid and open data networks to maintain a highly efficient, transparent, and eco-friendly urban ecosystem.

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Copenhagen, Denmark:

Focused on smart mobility and achieving zero emissions. The city relies on AI to control traffic lights based on bicycle traffic and congestion, monitors air quality in real-time to adjust municipal services, and operates district heating systems that recycle waste heat.

SMART CITIES

Barcelona, Spain:

A pioneer in smart street furniture. Barcelona uses "smart streetlights" equipped with sensors that adjust lighting based on pedestrian presence and monitor environmental noise and air quality. The city also uses smart sensors to direct drivers to available parking spaces, significantly reducing traffic congestion.

THANK YOU FOR YOUR ATTENTION!

SMART CITIES: MILAN

Read the following text about Milan, the leading Smart City in Italy.

Milan: smart mobility, digital twin and urban sustainability

Milan is the **leading Smart City in Italy**, with **numerous projects aimed** at **improving sustainability** and **urban innovation**. The **Smart City Lab** project—an R&D incubator—provides an environment where startups and companies can test innovative solutions in areas such as sustainable mobility, energy and the environment.

Within the **framework** of **NRPP funds** for digitalization and innovation in Public Administration, the **Municipality** of Milan has been **selected** to **experiment** with the **Mobility as a Service project**, which aims to promote greater accessibility, multimodality and sustainability in commuting by reducing the use of private transportation. In addition, there is **Smart Parking**, which uses IoT sensors to monitor parking availability and optimize urban traffic management.

Furthermore, the Lombard capital has launched the **ForestaMi** project, which aims to plant three million trees by 2030 to improve air quality and counteract pollution. Another initiative, launched at the **beginning of 2025**, is the **Smart City and Extended Digital Twin development project**, which, through a virtual replica of the city, aims to enhance urban planning and the management of public services.
Source:
https://www.intesasanpaoloinnovationcenter.com/en/news-and-events/news/2023/06/smart-city-where-we-are-in-italy/

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INTERESTING STATISTICS ABOUT SMART CITIES

- The global Smart Cities market size will grow from approximately US$ 410.8 billion in 2020 to US$ 820.7 billion by 2025.
- In 2008, about 50% of the world's population lived in cities. By 2040, this number is expected to increase to 65%.
- There are already more than 500 cities globally that have at least 1 million inhabitants. By way of comparison, in 1900, there were only 12 such cities.
- In 1975, there were only three megacities (+10 million citizens) around the globe: New York, Tokyo and Mexico City. Today, there are 21 megacities worldwide, and by 2025, there will be 29.
- There are four key areas of Smart City investment: innovation, buildings and construction, energy and water and waste management.
- The use of digital government services tripled between 2006 and 2016 in OECD countries, with 36% of citizens submitting forms through public authority websites in 2016.
- In 2020, Singapore was ranked as the smart city most ready to host autonomous vehicles.
- New York has installed cameras and sensors at 10,000 city intersections, aiming to invest in connected infrastructure and adaptive signals.
- In 2020, London launched the Smart Mobility Living Lab (SMLL), the world's most advanced urban testbed. SMLL provides 5G connectivity for connected and autonomous vehicles.
- Since 2010, since Paris first started implementing smart solutions to improve road safety and traffic management, traffic deaths in the city have fallen by 40%.

Source: https://neos.ro/en/10-statistics-to-know-about-smart-cities/

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SMART CITIES: WRITING

Which technologies do you find most useful related to the transformation of urban living? How do they improve the lives of citizens? Write 6-8 sentences.

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